Homework Template

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Problem 1 — Conventional Polynomial Evaluation

We are given the conventional algorithm:

procedure POLYNOMIAL($c, a_0, a_1, \ldots, a_n :$ real numbers)
$power \leftarrow 1$
$y \leftarrow a_0$
for $i = 1$ to n **do**
$power \leftarrow power \cdot c$
$y \leftarrow y + a_i \cdot power$
end for
return y

(a) Evaluate $3x^2 + x + 1$ at $x = 2$ by tracing each step

Here $n = 2$, $c = 2$, and $(a_0, a_1, a_2) = (1, 1, 3)$. The table logs every assignment step.

Step	Action	power	y
1	$power \leftarrow 1$	1	–
2	$y \leftarrow a_0$	1	1
3	$i = 1$: $power \leftarrow power \cdot c = 1 \cdot 2$	2	1
4	$i = 1$: $y \leftarrow y + a_1 \cdot power = 1 + 1 \cdot 2$	2	3
5	$i = 2$: $power \leftarrow power \cdot c = 2 \cdot 2$	4	3
6	$i = 2$: $y \leftarrow y + a_2 \cdot power = 3 + 3 \cdot 4$	4	15

Return: $y = 15$. Thus $3(2)^2 + 2 + 1 = 15$.

(b) Exact operation counts

Initialization ($power \leftarrow 1$, $y \leftarrow a_0$) performs no additions or multiplications.
Each loop iteration ($i = 1, \ldots, n$) performs:

- $power \leftarrow power \cdot c$: **1 multiplication**
- $y \leftarrow y + a_i \cdot power$: **1 multiplication** and **1 addition**

Therefore, over n iterations:

$$\boxed{\text{Multiplications} = 2n} \quad \text{and} \quad \boxed{\text{Additions} = n}.$$

Problem 2 — Horner's Method

There is a more efficient algorithm for evaluating polynomials than the conventional method from Problem 1. Horner's method evaluates

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

at $x = c$ using the pseudocode:

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procedure HORNER(c, a_0, a_1, ..., a_n : real numbers)
    y ← a_n
    for i = 1 to n do
        y ← y · c + a_{n−i}
    end for
    return y
end procedure
```

(a) Evaluate $3x^2 + x + 1$ at $x = 2$ by tracing each step

Here $n = 2$, $c = 2$, and $(a_0, a_1, a_2) = (1, 1, 3)$.

Step	Action	y
1	$y \leftarrow a_2 = 3$	3
2	$i = 1$: $y \leftarrow y \cdot c + a_{2-1} = 3 \cdot 2 + 1$	7
3	$i = 2$: $y \leftarrow y \cdot c + a_{2-2} = 7 \cdot 2 + 1$	15

Return: $y = 15$. Thus $3(2)^2 + 2 + 1 = 15$.

(b) Exact operation counts

Each loop iteration performs:

- one multiplication: $y \cdot c$,
- one addition: $+\,a_{n-i}$.

With n iterations total, the exact counts are:

$$\boxed{\text{Multiplications} = n} \qquad \boxed{\text{Additions} = n}.$$

Problem 3 — Prove $f(x) = O(g(x))$ by definition

(a) $x^4 + 9x^3 + 4x + 7$ is $O(x^4)$

For $x \geq 1$,

$$x^4 + 9x^3 + 4x + 7 \leq x^4 + 9x^4 + 4x^4 + 7x^4 = 21x^4.$$

Hence with $c = 21$ and $x_0 = 1$, we have $f(x) \leq c\,x^4$ for all $x \geq x_0$, so $f(x) = O(x^4)$.

(b) $2^x + 17$ is $O(3^x)$

For all $x \geq 0$, $2^x \leq 3^x$ and $17 \leq 17 \cdot 3^x$. Thus

$$2^x + 17 \leq 3^x + 17 \cdot 3^x = 18 \cdot 3^x.$$

With $c = 18$ and $x_0 = 0$, $2^x + 17 = O(3^x)$.

(c) $\dfrac{x^2+1}{x+1}$ is $O(x)$

For $x \geq 1$ we have $x+1 \geq x$ and $x^2+1 \leq 2x^2$, so

$$\frac{x^2+1}{x+1} \leq \frac{2x^2}{x} = 2x.$$

With $c = 2$ and $x_0 = 1$, the function is $O(x)$.

(d) For each fixed $k \in \mathbb{Z}_{>0}$, $1^k + 2^k + \cdots + n^k$ is $O(n^{k+1})$

Each term satisfies $i^k \leq n^k$ for $1 \leq i \leq n$, so

$$\sum_{i=1}^{n} i^k \leq n \cdot n^k = n^{k+1}.$$

Thus with $c = 1$ and $n_0 = 1$, the sum is $O(n^{k+1})$ for every fixed k.

Problem 4 — Recursive exponentiation

We compute a^n by the recurrence: let $h = a^{\lfloor n/2 \rfloor}$; if n is even return $h \cdot h$, else return $a \cdot h \cdot h$, and for $n = 0$ return 1. Assume unit-cost arithmetic.

(a) Worst-case time recurrence $T(n)$

There is one recursive call on $\lfloor n/2 \rfloor$ and $O(1)$ extra work (a constant number of multiplications and parity check), so

$$T(0) = \Theta(1), \qquad T(n) = T(\lfloor n/2 \rfloor) + \Theta(1) \quad \text{for } n > 0.$$

(b) Solve via the Master Theorem

This is the case $a = 1$, $b = 2$, $f(n) = \Theta(1)$. Since $n^{\log_b a} = n^0 = 1$ and $f(n) = \Theta(1)$, it is the balanced case:

$$T(n) = \Theta(\log n).$$

Thus exponentiation by repeated squaring runs in $\Theta(\log n)$ time.

Problem 5 — Counting with generating functions

Setup. We use ordinary generating functions (OGFs) for identical items and coefficient extraction $[z^m]F(z)$.

(i) 10 identical balloons to 4 children, each at least 2

Each child has OGF $z^2/(1-z)$. Total OGF $\left(\dfrac{z^2}{1-z}\right)^4 = z^8(1-z)^{-4}$. We need $[z^{10}]$, i.e., $[z^2](1-z)^{-4} = \binom{2+4-1}{4-1} = \binom{5}{3} = 10$.

$$\boxed{10}$$

(ii) 12 identical action figures to 5 children, each at most 3

Each child has OGF $1 + z + z^2 + z^3$. Total OGF $(1 + z + z^2 + z^3)^5$. We need the coefficient of z^{12}:

$$[z^{12}](1 + z + z^2 + z^3)^5 = \boxed{35}.$$

(iii) 15 identical stuffed animals to 6 children, each between 1 and 3

Each child has OGF $z + z^2 + z^3 = z(1 + z + z^2)$. Total OGF $z^6(1 + z + z^2)^6$. We need $[z^{15}]$, i.e., $[z^9](1 + z + z^2)^6 = \boxed{50}$.

(iv) Dozen bagels from egg, salty, plain; at least 2 of each, and at most 3 salty

Egg and plain each: $z^2/(1 - z)$; salty: $z^2 + z^3$. Total OGF

$$\frac{z^2}{1-z} \cdot \frac{z^2}{1-z} \cdot (z^2 + z^3) = z^6 \frac{1+z}{(1-z)^2}.$$

We need $[z^{12}]$, i.e., $[z^6]\dfrac{1+z}{(1-z)^2}$. Since $(1 - z)^{-2} = \sum_{n\geq 0}(n + 1)z^n$, we get

$$[z^6]\frac{1+z}{(1-z)^2} = (6 + 1) + (5 + 1) = \boxed{13}.$$

Problem 6 — Generating functions in closed form

Let $G(z) = \sum_{n\geq 0} z^n = \frac{1}{1-z}$ for $|z| < 1$. Then

$$\sum_{n\geq 0} nz^n = zG'(z) = \frac{z}{(1-z)^2}, \qquad \sum_{n\geq 0} n^2 z^n = z\frac{d}{dz}\left(\frac{z}{(1-z)^2}\right) = \frac{z(1+z)}{(1-z)^3}.$$

(a) $a_n = n$

$$A(z) = \sum_{n\geq 0} nz^n = \boxed{\frac{z}{(1-z)^2}}.$$

(b) $a_n = 2n + 3$

$$A(z) = 2\sum_{n\geq 0} nz^n + 3\sum_{n\geq 0} z^n = 2 \cdot \frac{z}{(1-z)^2} + 3 \cdot \frac{1}{1-z} = \boxed{\frac{2z}{(1-z)^2} + \frac{3}{1-z}}.$$

(c) $a_n = n^2$

$$A(z) = \sum_{n\geq 0} n^2 z^n = \boxed{\frac{z(1+z)}{(1-z)^3}}.$$

(d) $a_n = 2n^2 + 3n + 4$

$$\begin{aligned} A(z) &= 2\sum_{n\geq 0} n^2 z^n + 3\sum_{n\geq 0} nz^n + 4\sum_{n\geq 0} z^n \\ &= 2 \cdot \frac{z(1+z)}{(1-z)^3} + 3 \cdot \frac{z}{(1-z)^2} + 4 \cdot \frac{1}{1-z} \\ &= \boxed{\frac{2z(1+z)}{(1-z)^3} + \frac{3z}{(1-z)^2} + \frac{4}{1-z}}. \end{aligned}$$

References